

Primordial Non-Gaussianity

John Joseph Marchetta

Research Team:

Dr. Cleaver (Advisor), Rui Pan, Jamal
Saeed, Dr. Wang



Outline

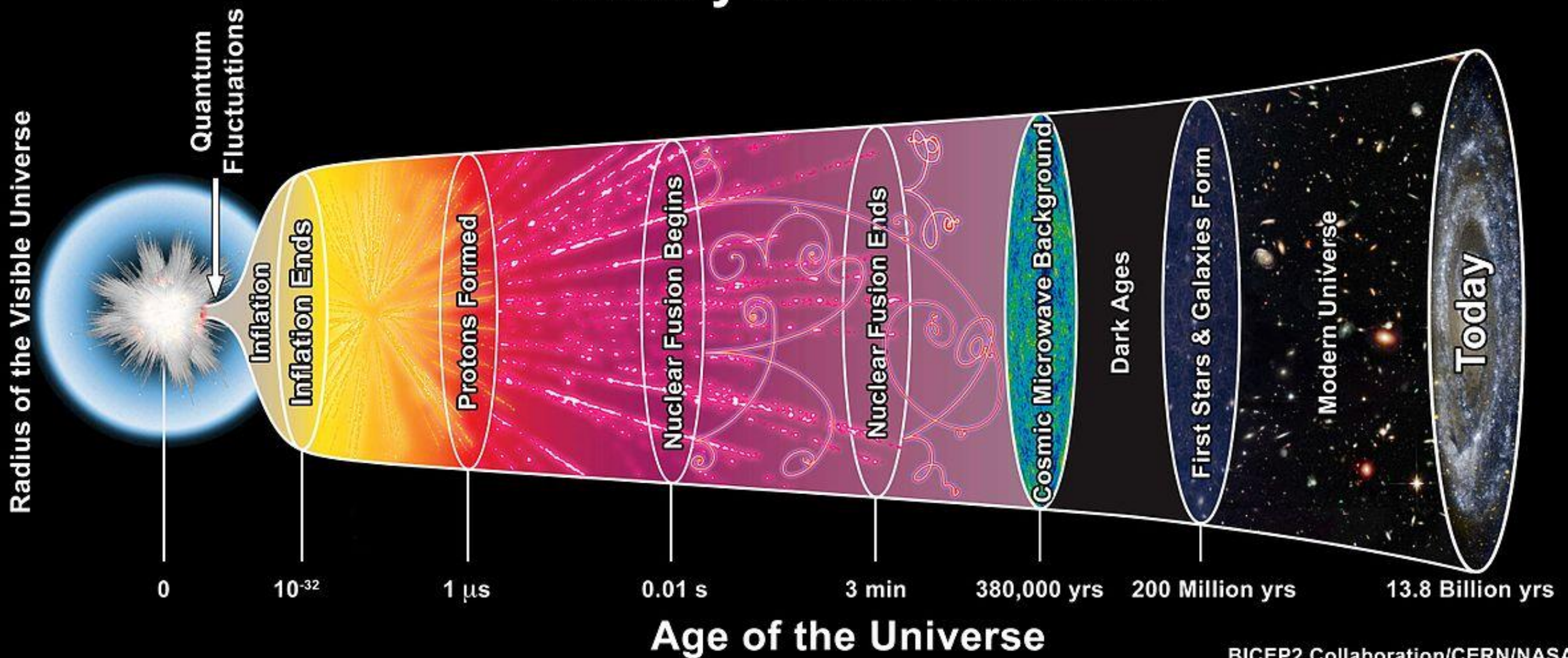
Early Universe

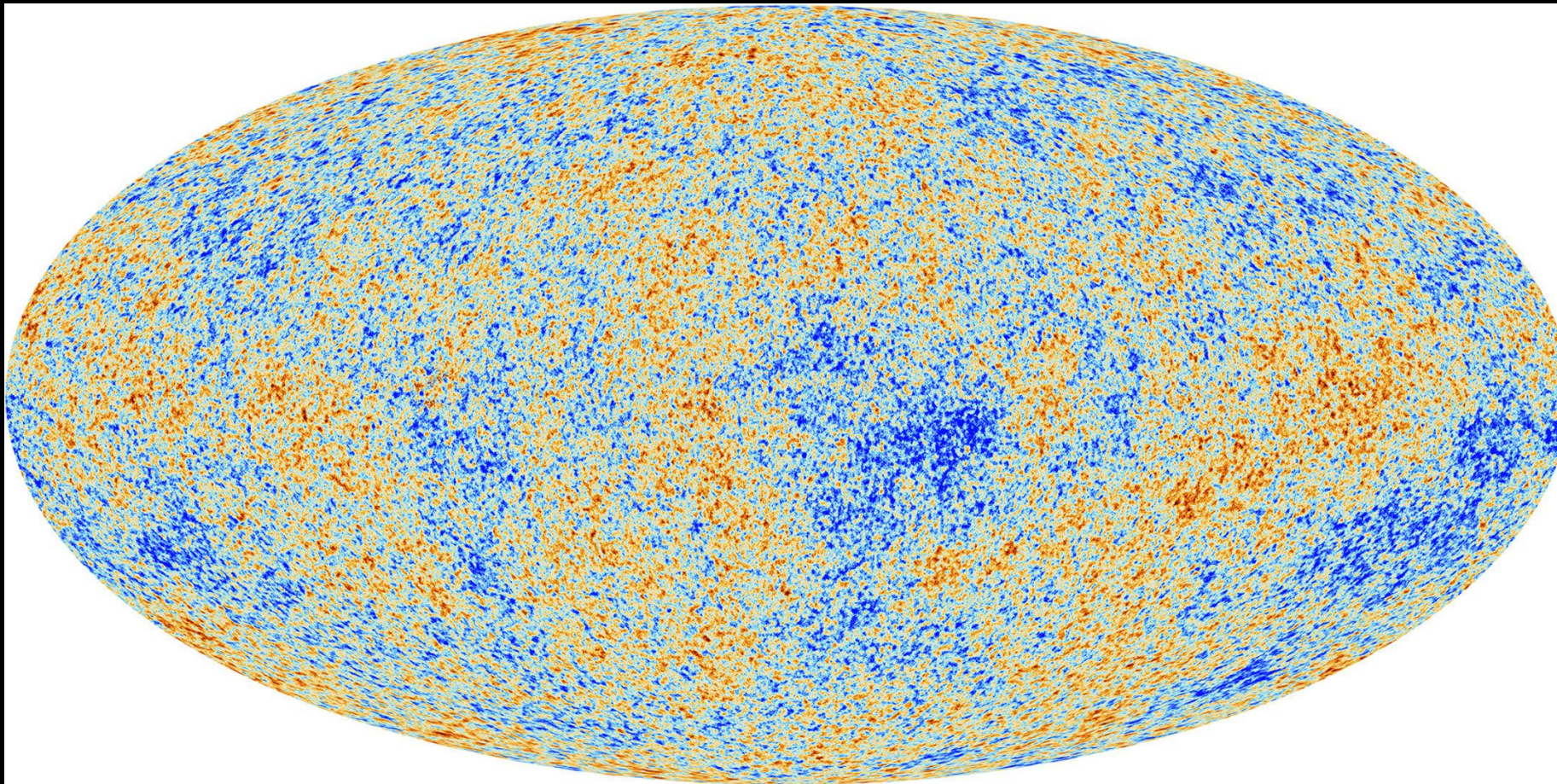
(Non)/gaussianity

Formalism

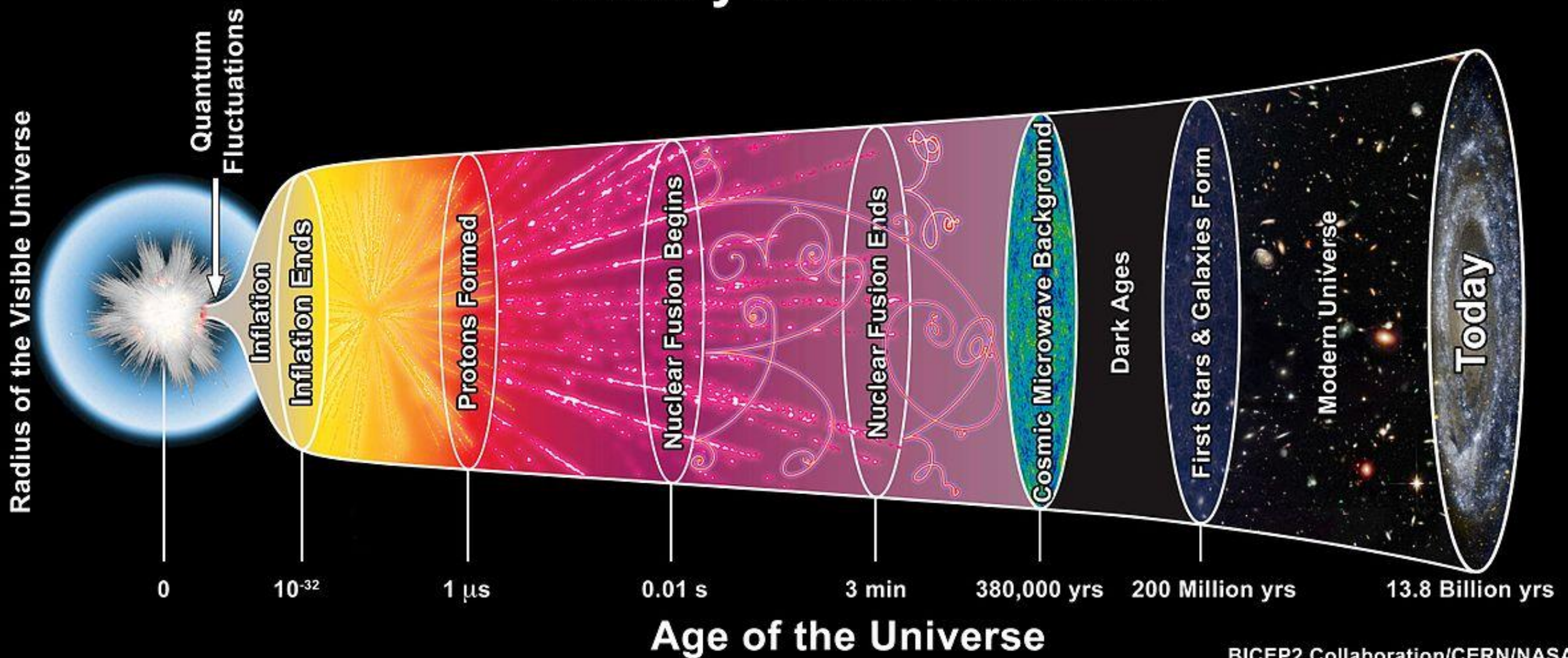
Loop quantum cosmology

History of the Universe

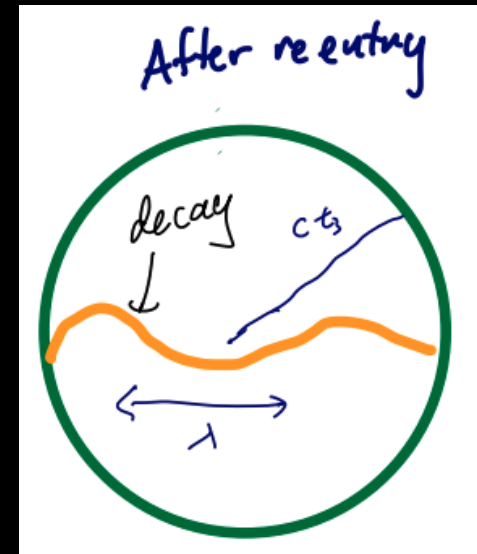
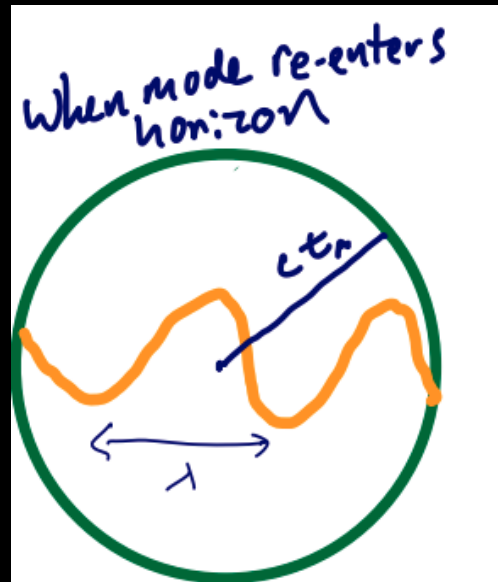
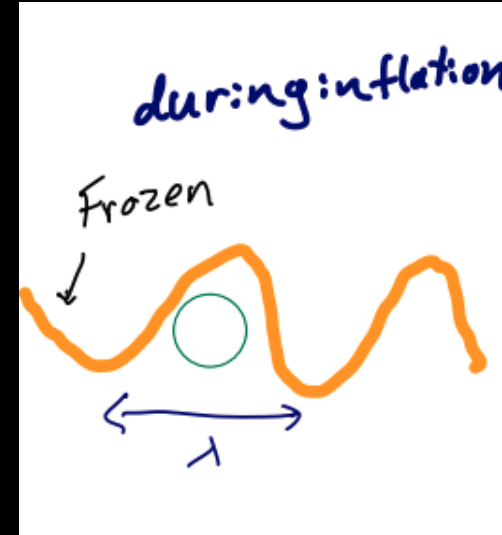
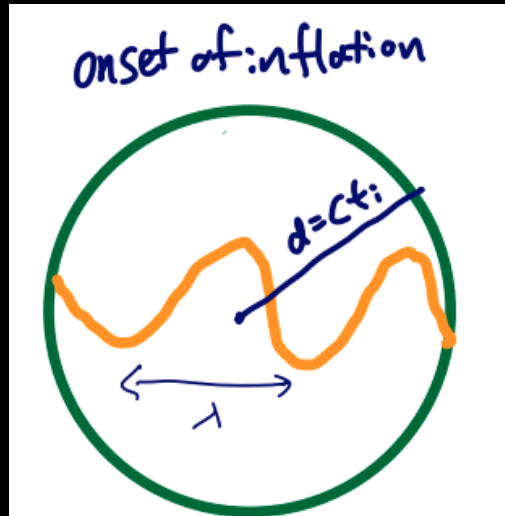




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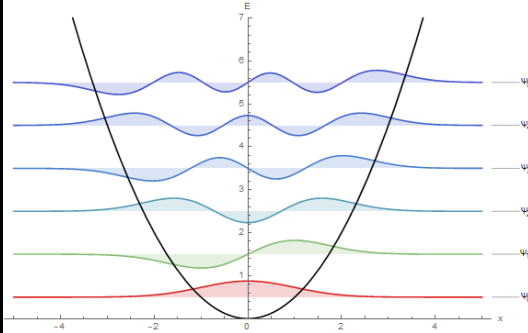


Physics of early universe perturbations



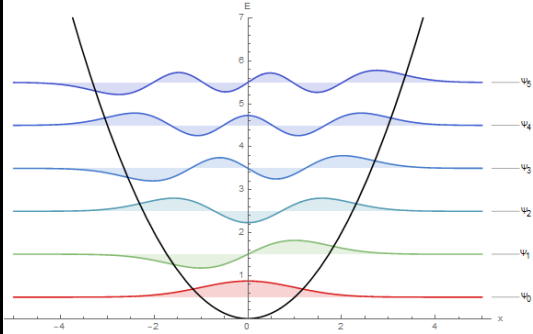
Review

- Harmonic Oscillator:



Review

- Harmonic Oscillator:



- Quantum Field Theory:

Ground State?

Side Notes:

Particles are excitation

$$a_{k2}^\dagger |0, 0, 0.. > = |0, 1, 0... >$$

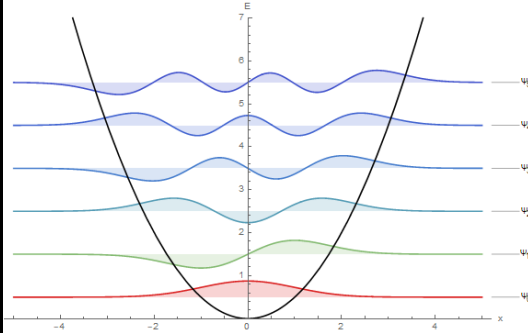
$$a_{k2} |0, 1, 0.. > = |0, 0, 0... >$$

Vacuum=> Infinite energy

$$\sum_{k=1}^{\infty} h\omega_k (n + \frac{1}{2}) = \sum_{k=1}^{\infty} \frac{h\omega_k}{2} = \infty$$

Review

- Harmonic Oscillator:

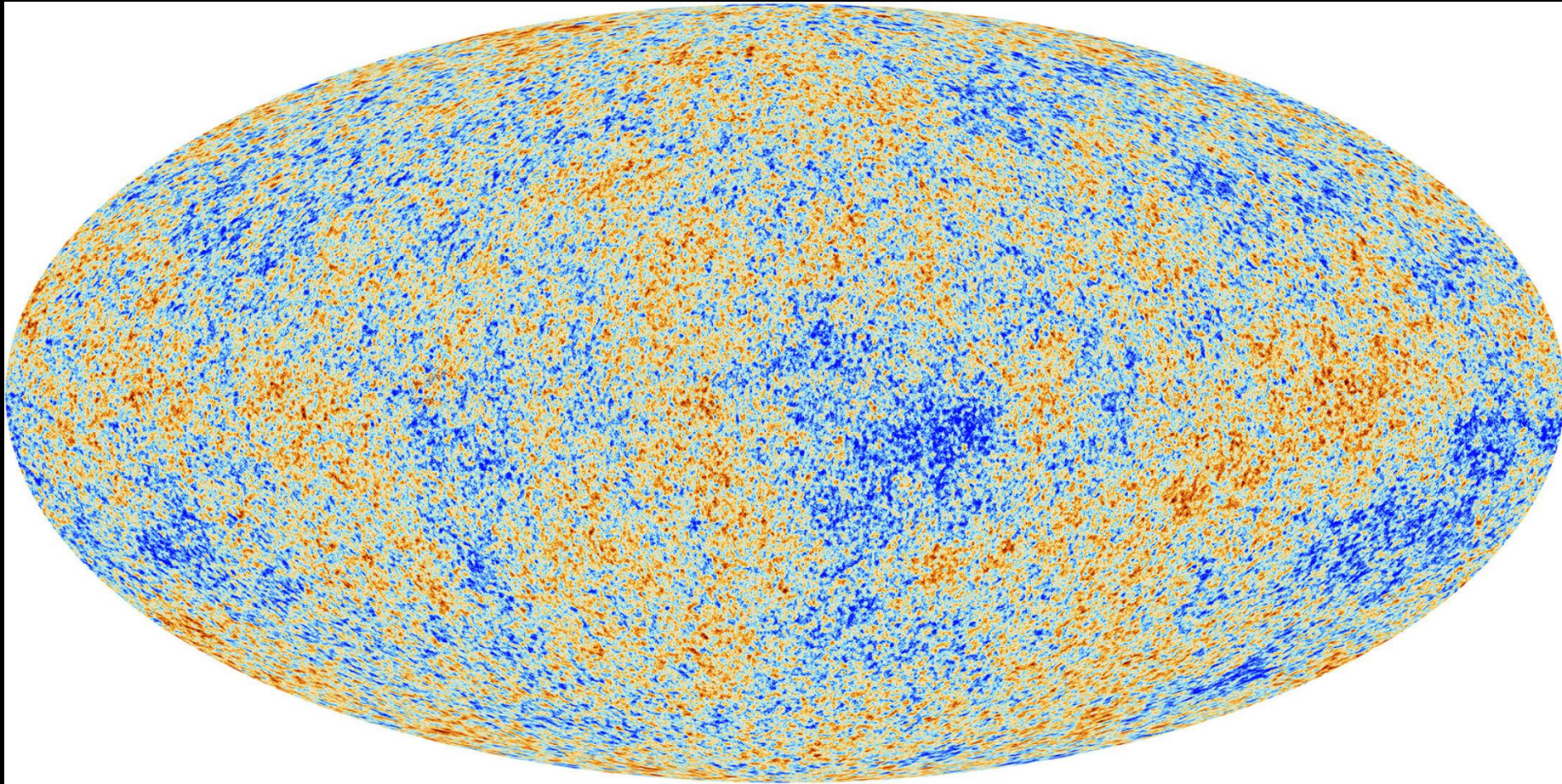


- Quantum Field Theory:

Ground State?

$$u_0(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-m\omega x^2 / 2\hbar}$$

$$\begin{aligned} \Psi_0[\phi] &\propto \exp\left(-\frac{1}{2\hbar} \int d^3k \sqrt{\mathbf{k}^2 + m^2} |\phi_{\mathbf{k}}|^2\right) \\ &\propto \prod_{\mathbf{k} \in \mathbb{R}^3} \exp\left(-\frac{1}{2\hbar} \sqrt{\mathbf{k}^2 + m^2} |\phi_{\mathbf{k}}|^2\right). \end{aligned}$$



Gaussian!

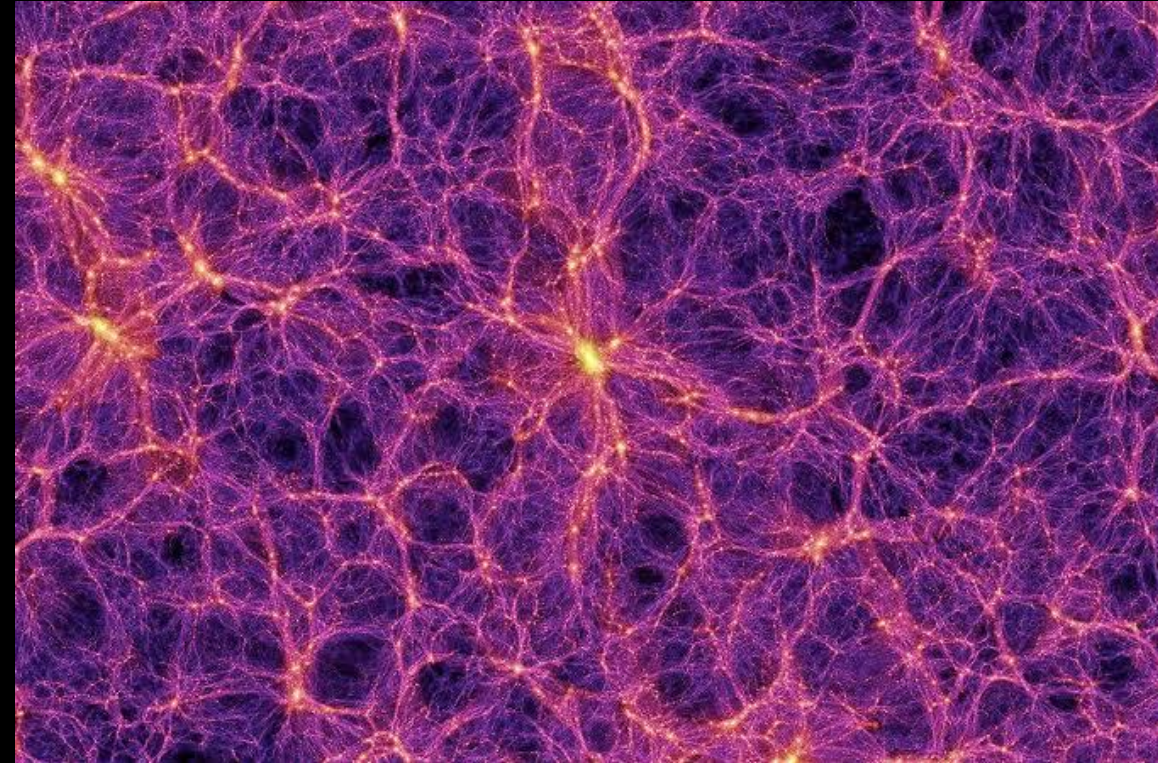
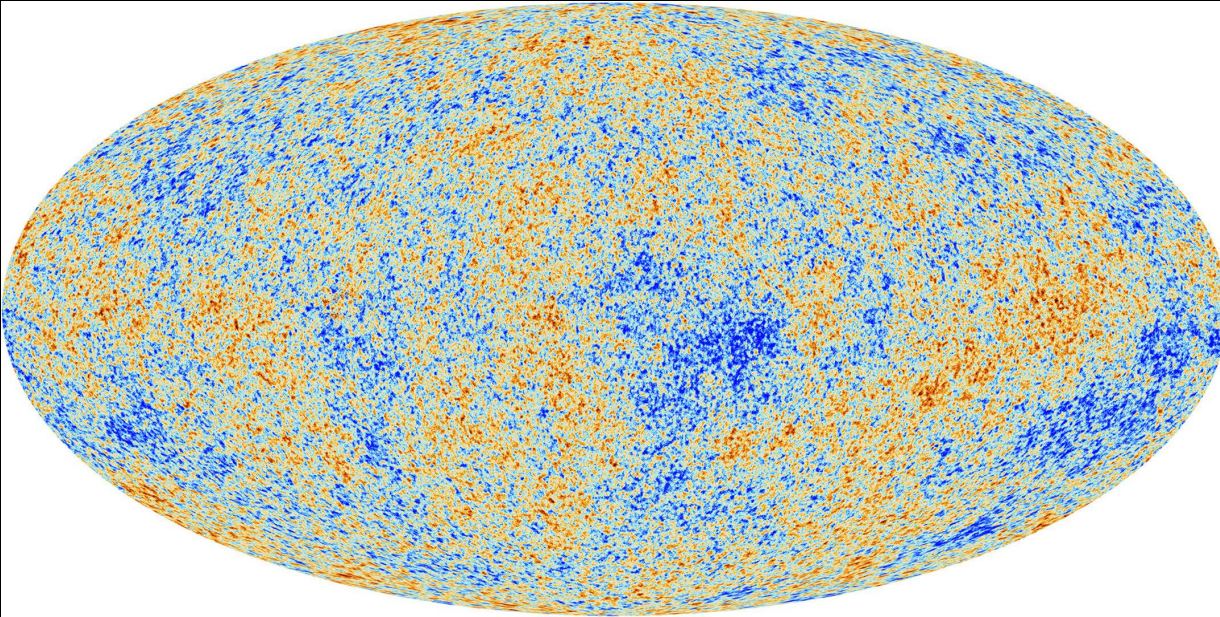
$$\langle \zeta(k_1)\zeta(k_2)\zeta(k_3) \rangle = 0$$

How we determine Quantum Field
Interactions?

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How we determine Quantum Field Interactions in the early Universe?



- Power Spectrum:

$$S = \int d^4x \sqrt{g} \left[\frac{M_P^2 R}{2} - \frac{1}{2} (\nabla \phi)^2 - V(\phi) \right]$$

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Coord. Choice: $\delta\phi = 0$

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$$[a, a^\dagger] = 1$$

- Power Spectrum (Int. Pic):

$$\langle 0 | \zeta_{k1}^I \zeta_{k2}^I | 0 \rangle = \langle 0 | (\zeta_{k1}^{I+} + \zeta_{k1}^{I-})(\zeta_{k2}^{I+} + \zeta_{k2}^{I-}) | 0 \rangle = \frac{H^2}{2(2\pi)^3 \epsilon k_1^3} \delta(k_1 - k_2)$$

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- Expand action to third order in pert + local field re-def (results):

$$S_3 = \int \frac{\dot{\phi}^4}{\dot{\rho}^4} e^{5\rho} \dot{\rho} \dot{\zeta}_c^2 \partial^{-2} \dot{\zeta}_c + \dots$$

$$\zeta_I \rightarrow R_g$$

$$\mathcal{R}(\mathbf{x}) = \mathcal{R}_g(\mathbf{x}) + \frac{3}{5} f_{\text{NL}}^{\text{local}} [\mathcal{R}_g(\mathbf{x})^2 - \langle \mathcal{R}_g(\mathbf{x})^2 \rangle]$$

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- Bi-Spectrum:

$$\langle \mathcal{R}_{\mathbf{k}_1} \mathcal{R}_{\mathbf{k}_2} \mathcal{R}_{\mathbf{k}_3} \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{\mathcal{R}}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3).$$

$$B_{\mathcal{R}}(k_1, k_2, k_3) = \frac{6}{5} f_{\text{NL}}^{\text{local}} \times [P_{\mathcal{R}}(k_1) P_{\mathcal{R}}(k_2) + P_{\mathcal{R}}(k_2) P_{\mathcal{R}}(k_3) + P_{\mathcal{R}}(k_3) P_{\mathcal{R}}(k_1)]$$

In-In formalism (Sakurai Chapter 5)

$|0 \text{ in} \rangle$ Vacuum at $t=-\infty$

$|0 \text{ out} \rangle$ Vacuum at $t= \infty$

$$H_{int} = -L_{int}$$

$$\langle in | R R R | in \rangle = \langle in | \left[\bar{T} e^{(i \int_{-\infty(1-i\epsilon)}^t H_{int}^3(t') dt')} \right] R^I R^I R^I \left[T e^{(-i \int_{-\infty(1+i\epsilon)}^t H_{int}^3(t') dt')} \right] | in \rangle$$

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$$\langle in | RRR | in \rangle = \langle in | (1 + i \int_{-\infty(1-i\epsilon)}^t H_{int}^3(t') dt') R^I R^I R^I (1 - i \int_{-\infty(1+i\epsilon)}^t H_{int}^3(t') dt') | in \rangle$$

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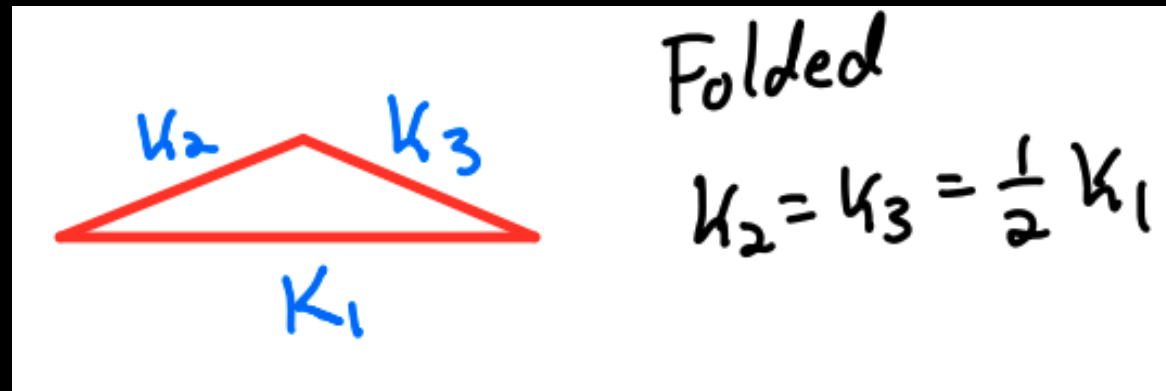
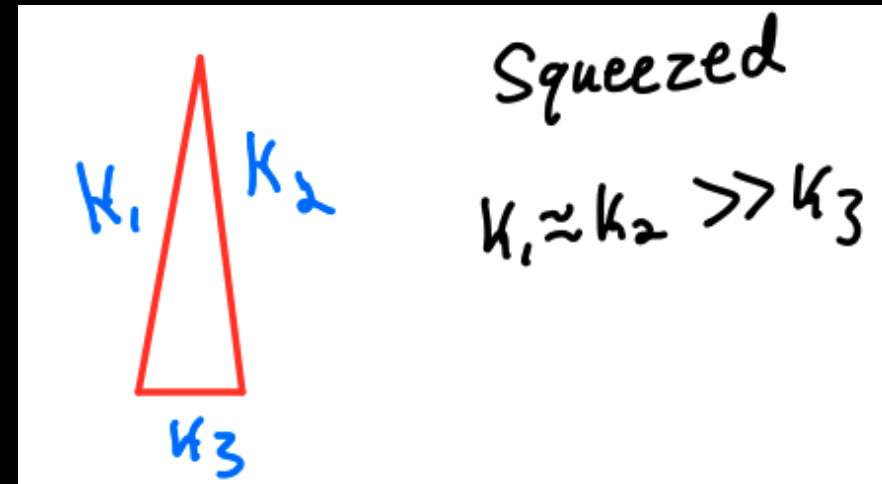
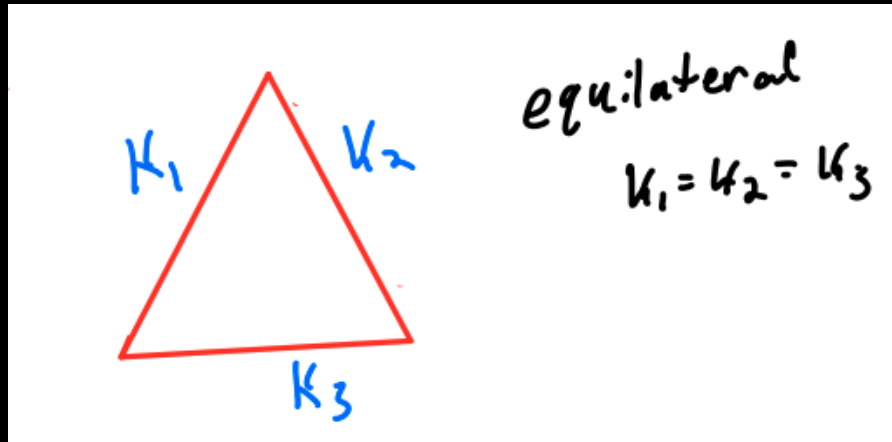
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- Thus solve for:

$$f_{NL}^{\text{local}}$$

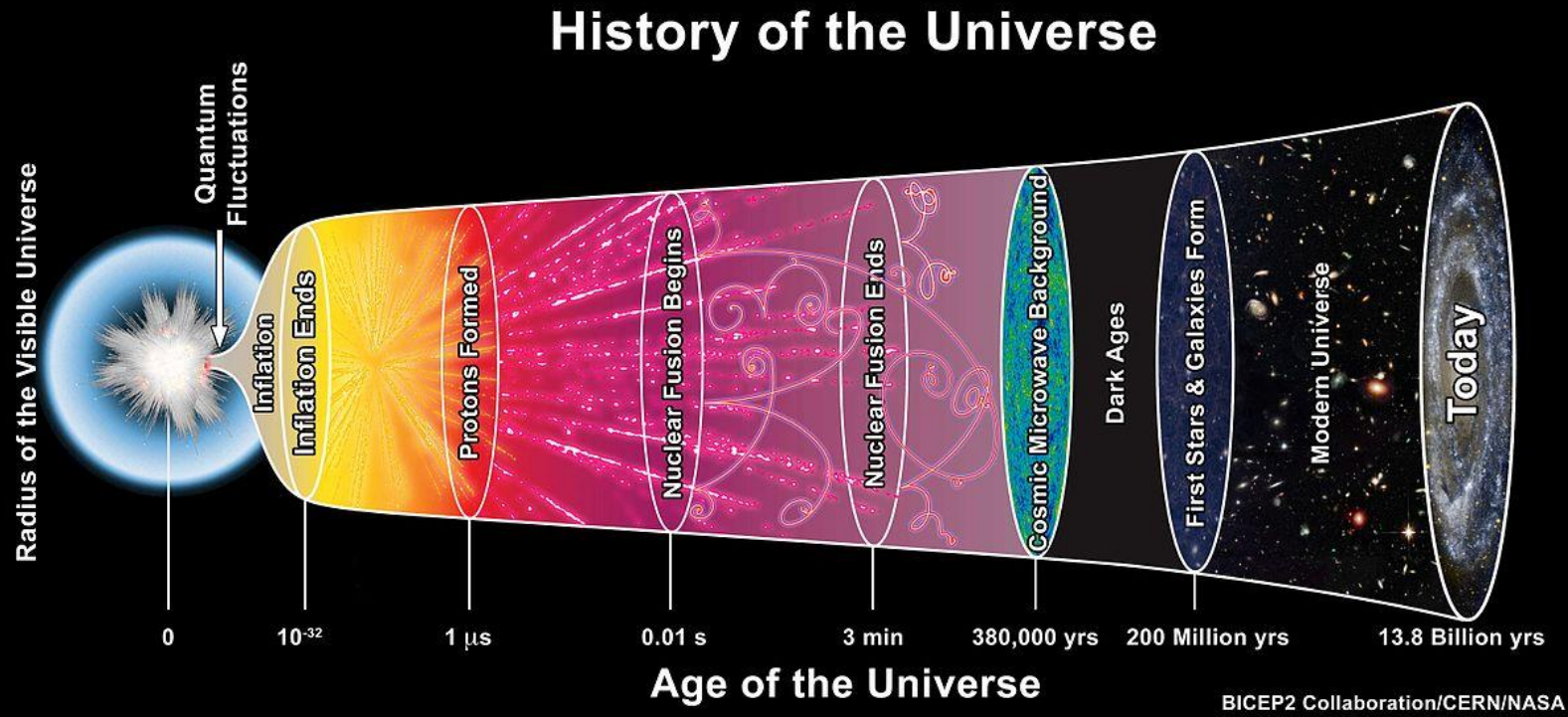
Shapes

- Different theory predicts different peak shapes, great to distinguish between models



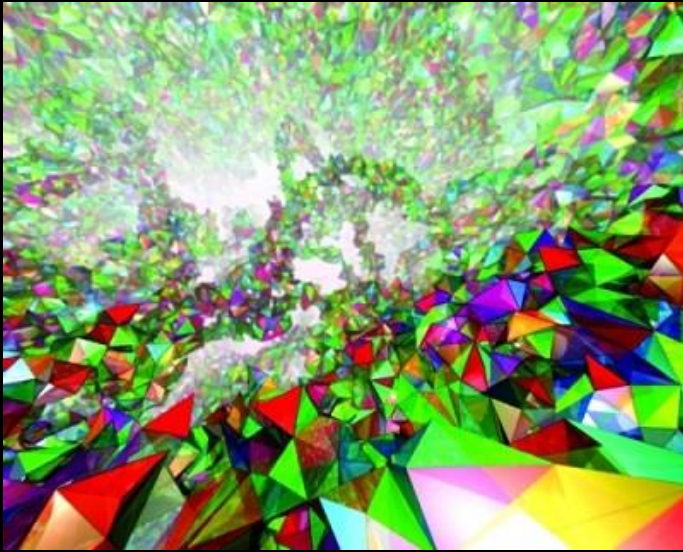
A way to get large non-gaussianity

A way is to look toward quantum gravity effects:

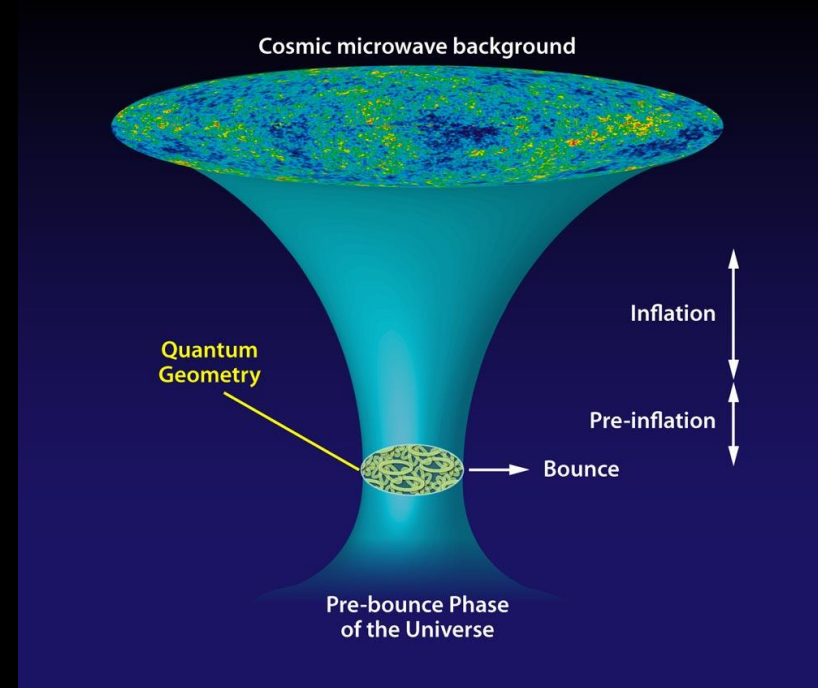


Loop quantum gravity/cosmology

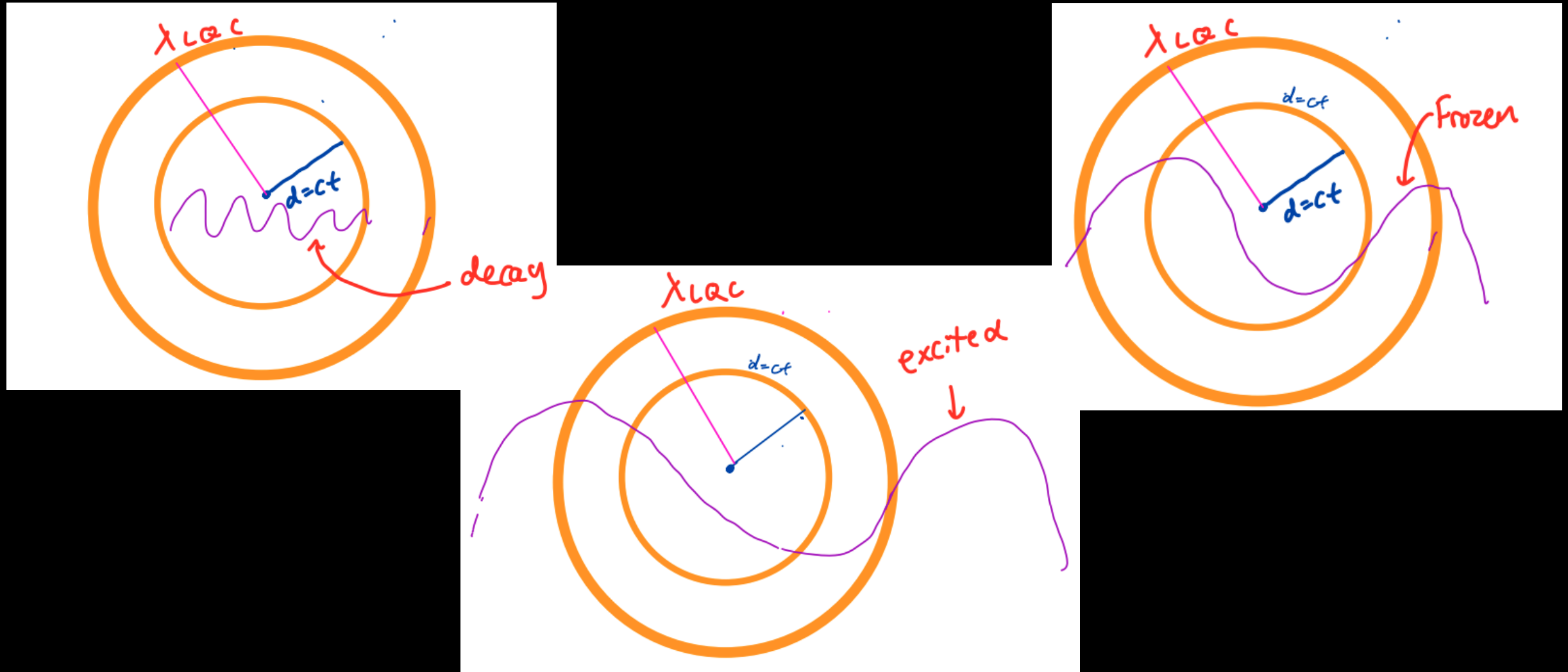
- Minimum area



- Bouncing Universe



How does LQC affect perturbations?



LQC modifications => Enhanced non-gaussianity

- "Primordial non-Gaussianity and power asymmetry with quantum gravitational effects in loop quantum cosmology" 2014. T.Zhu, A. Wang , K.Kirsten, G. Cleaver, and Q Sheng.

$$\langle \mathcal{R}_{\mathbf{k}_1} \mathcal{R}_{\mathbf{k}_2} \mathcal{R}_{\mathbf{k}_3} \rangle \simeq -i \left\langle \left[\mathcal{R}_{\mathbf{k}_1} \mathcal{R}_{\mathbf{k}_2} \mathcal{R}_{\mathbf{k}_3}, \int_{t_0}^t dt' H_3(t') \right] \right\rangle$$

$$H_3(t) = - \int d^3x \left\{ a^3 \epsilon_1^2 \mathcal{R} \dot{\mathcal{R}}^2 + a \epsilon_1^2 \mathcal{R} (\partial \mathcal{R})^2 \right. \\ \left. - 2a^3 \epsilon_1^2 \dot{\mathcal{R}} \partial \mathcal{R} \partial (\partial^{-2} \dot{\mathcal{R}}) \right. \\ \left. + \partial_t \left(-\frac{\epsilon_1 \epsilon_2}{2} a^3 \mathcal{R}^2 \dot{\mathcal{R}} \right) \right\}.$$

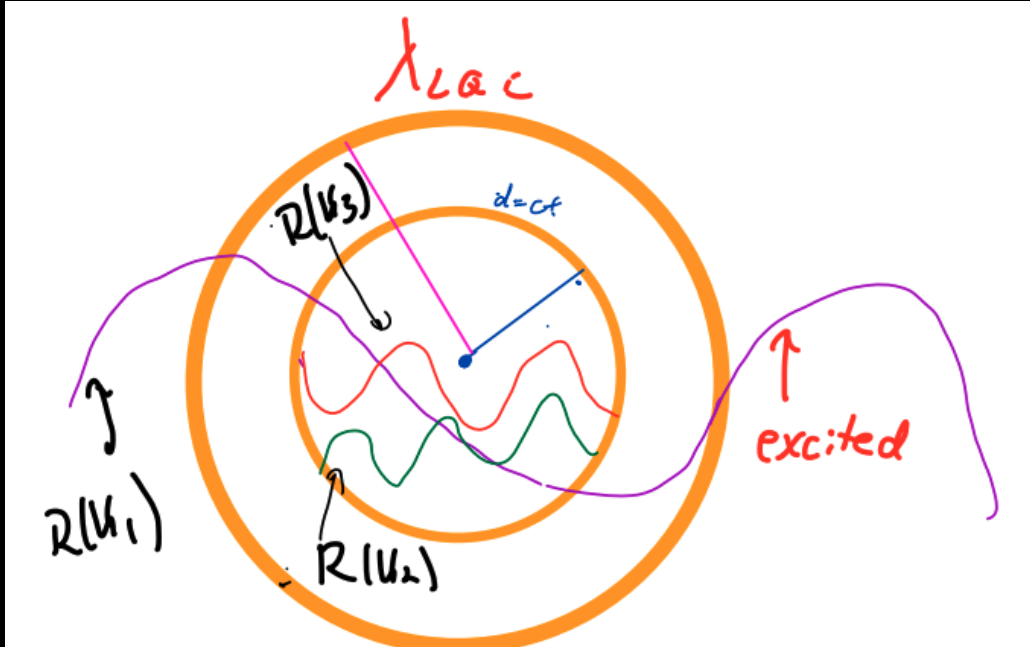
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Long perturbations enter inflation in a non standard vacuum=> large non-gaussianity in squeezed limit



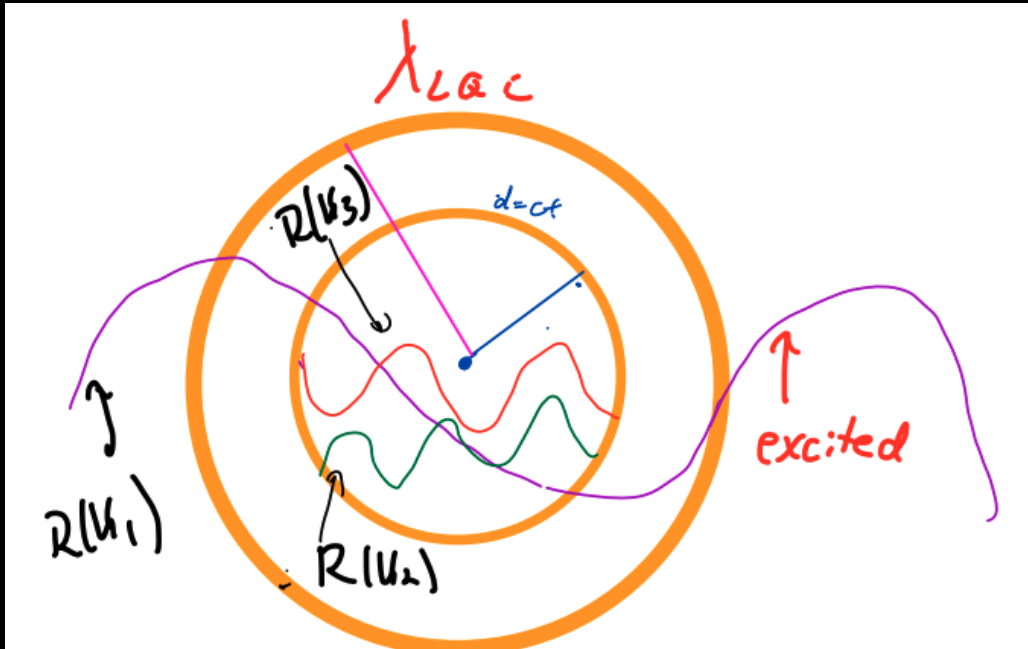
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Long perturbations enter inflation in a non standard vacuum=> large non-gaussianity in squeezed shape

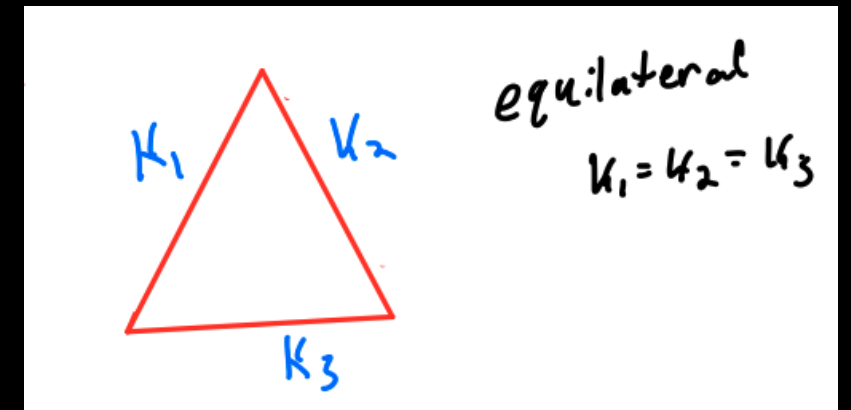
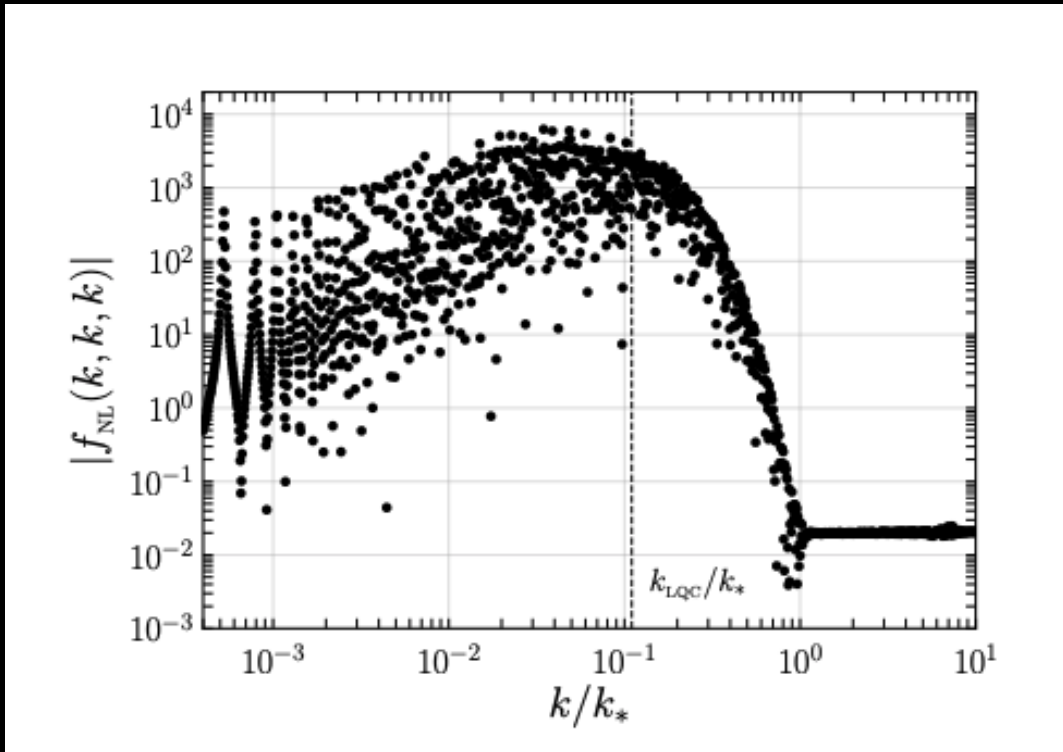


Scale dependent non-Gaussianity

This also can account for power asymmetry anomaly in CMB!!

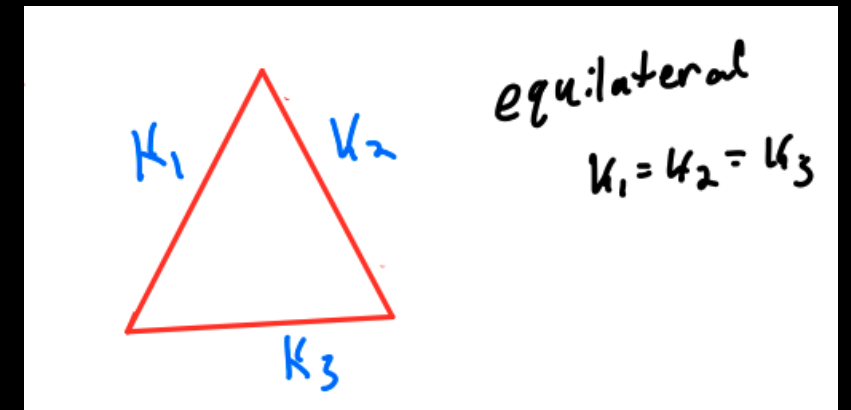
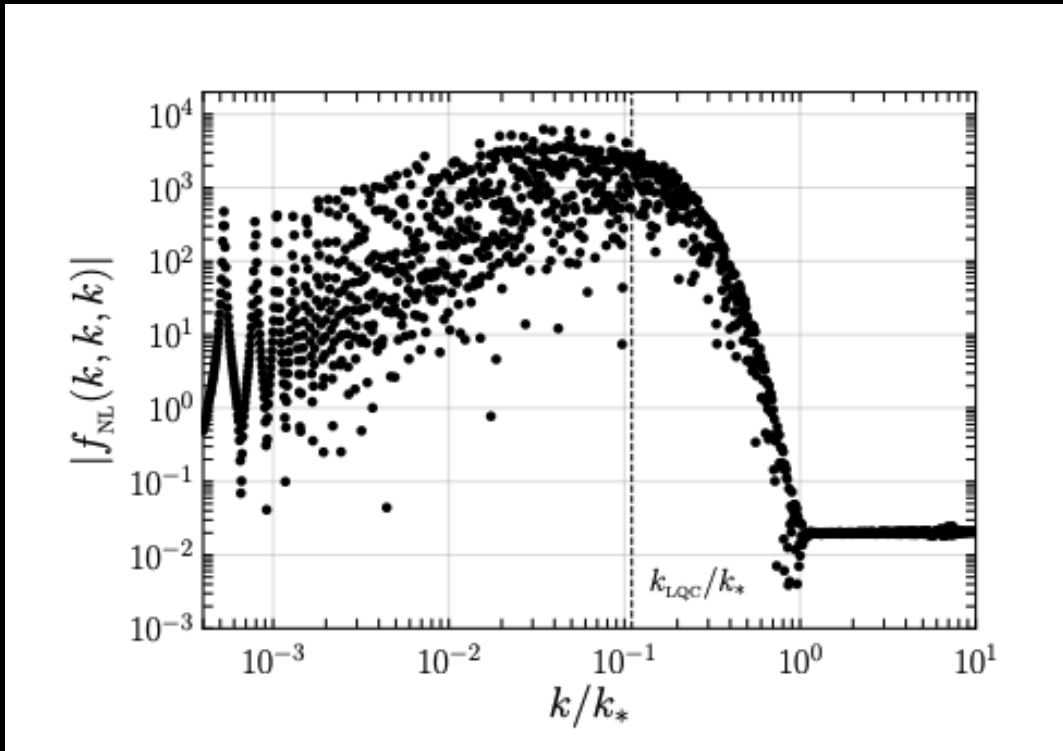
Interactions through the bounce

- Very sensitive to ρ_b t_0
- Equilateral:



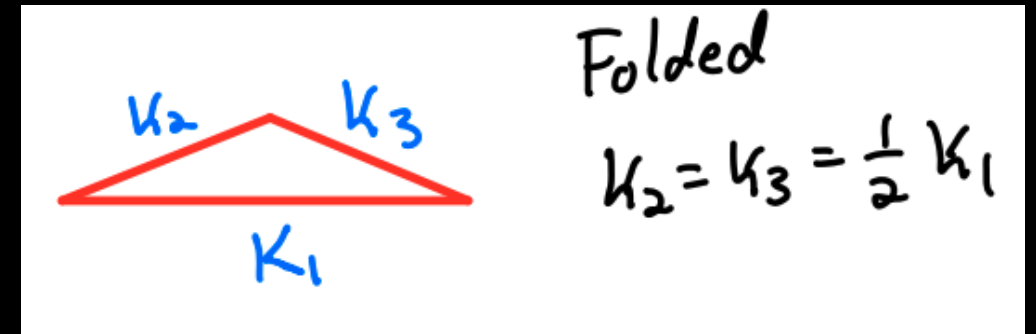
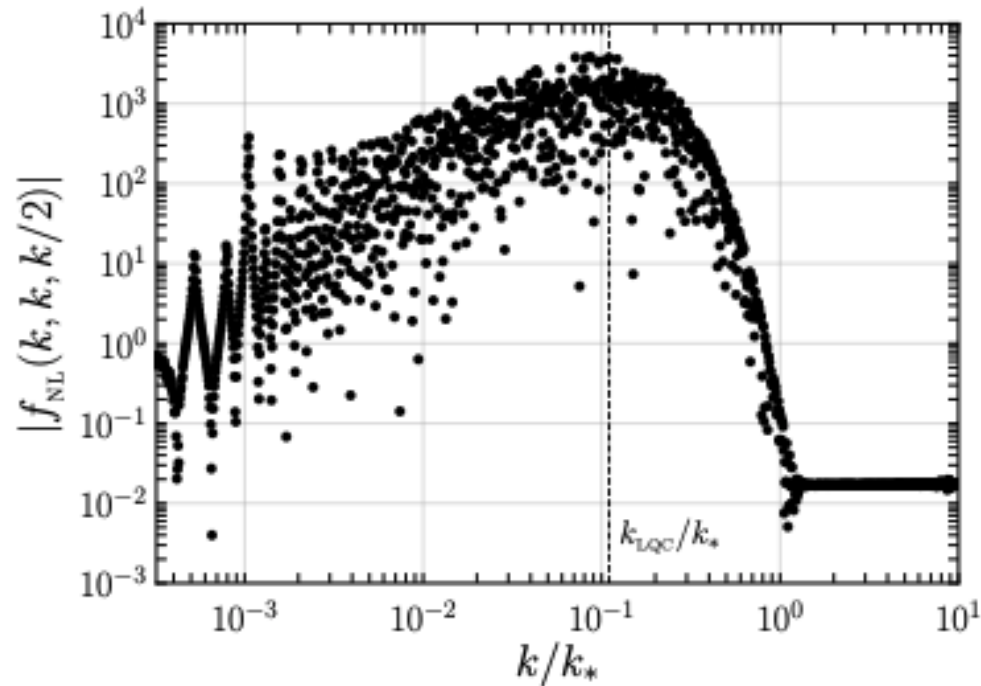
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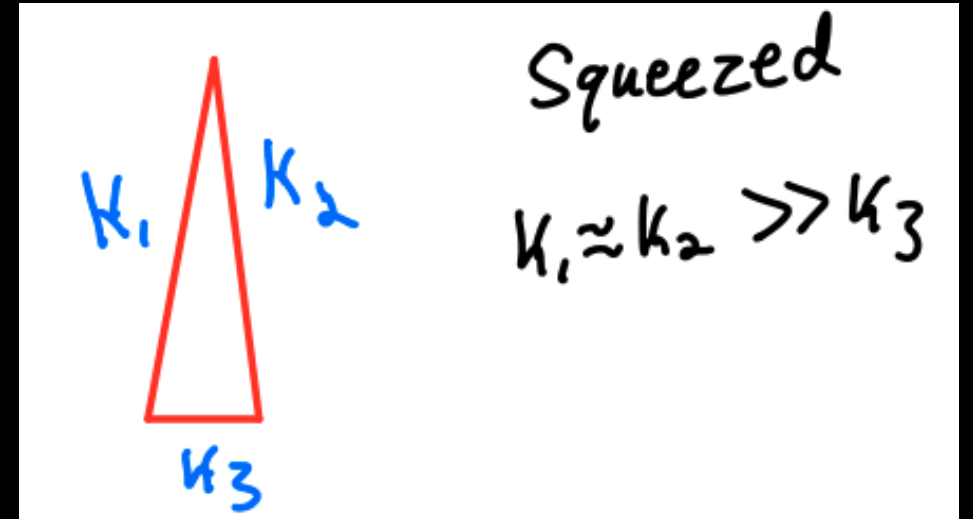
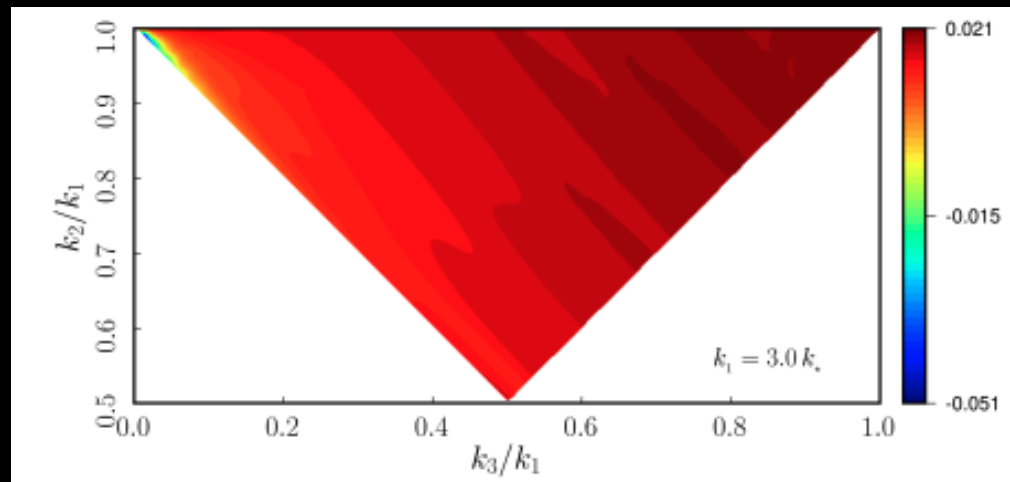
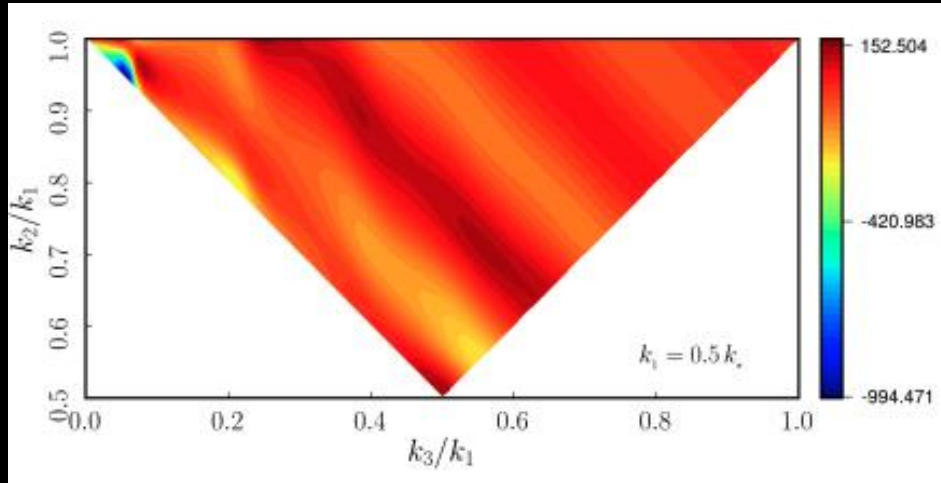
Interactions through the bounce

- Folded:



Interactions through the bounce

- Squeezed:



Conclusion

- Inflationary models generically predict a gaussian power spectrum in the CMB.
- Detection of non-gaussian features in the CMB, or primordial non-gaussian features in the large scale structure will be key in determining which model predicts the earliest conceived dynamics of the universe
- Loop quantum cosmology in a recipe which produces large non-gaussianities while retaining the desirable slow roll single field inflation and relieves CMB anomalies in the process.

Hope you had:

THANK YOU!!!!

Questions?



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$$\zeta_{\vec{k}}(t) = \zeta_k^{cl}(t) a_{\vec{k}}^\dagger + \zeta_k^{cl*}(t) a_{-\vec{k}}$$

- Power Spectrum (Int. Pic):

$$\langle 0 | \zeta_I(\mathbf{k}_1) \zeta_I(\mathbf{k}_2) | 0 \rangle = \langle 0 | (\zeta_I^+(\mathbf{k}_1) + \zeta_I^-(\mathbf{k}_1)) (\zeta_I^+(\mathbf{k}_2) + \zeta_I^-(\mathbf{k}_2)) | 0 \rangle = \frac{H^2}{2(2\pi)^3 \epsilon k_1^3} \delta(\mathbf{k}_1 - \mathbf{k}_2)$$

- Expanding to third order:
- Bi-Spectrum:

Formalism

- Assumption: Break fields into classical background (expectation value) and quantum perturbations.
- In

In-In formalism

- Vacuum fluctuations do not contribute as a contraction.

- Power spectrum:

$$\langle \mathcal{R}_{\mathbf{k}_1} \mathcal{R}_{\mathbf{k}_2} \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2) P_{\mathcal{R}}(k_1).$$

- Bi-spectrum only 2 indep modes. (Gaussianity disappears in 3 pt):

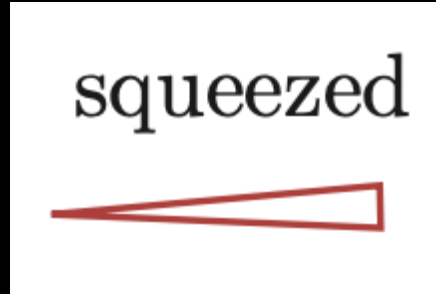
$$\langle \mathcal{R}_{\mathbf{k}_1} \mathcal{R}_{\mathbf{k}_2} \mathcal{R}_{\mathbf{k}_3} \rangle = (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) B_{\mathcal{R}}(\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3).$$

- Local NG:

$$B_{\mathcal{R}}(k_1, k_2, k_3) = \frac{6}{5} f_{\text{NL}}^{\text{local}} \times [P_{\mathcal{R}}(k_1)P_{\mathcal{R}}(k_2) + P_{\mathcal{R}}(k_2)P_{\mathcal{R}}(k_3) + P_{\mathcal{R}}(k_3)P_{\mathcal{R}}(k_1)]$$

Shapes

The delta function needs to satisfy the triangle inequality:



Local NG is largest in the squeezed limit.



Maldenca theorem

Maldenca calc

- Nongaussian correlations only depend on the time the perturbations exit the horizon.
- The loop contributions depend on the whole history of the evolution

- Starting point:

$$S = \frac{1}{2} \int \sqrt{g} [R - (\nabla \phi)^2 - 2V(\phi)]$$

- E.O.M:

$$3\dot{\rho}^2 = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$

$$\ddot{\rho} = -\frac{1}{2}\dot{\phi}^2$$

$$0 = \ddot{\phi} + 3\dot{\rho}\dot{\phi} + V'(\phi)$$

Weinberg Calc

- Going from 2nd to 3rd order formalism becomes tedious, so we introduce an "In-in formalism"
- Calculating the evolution from a time-dependent potential

$$\langle Q(t) \rangle = \sum_{N=0}^{\infty} i^N \int_{-\infty}^t dt_N \int_{-\infty}^{t_N} dt_{N-1} \cdots \int_{-\infty}^{t_2} dt_1 \langle [H_I(t_1), [H_I(t_2), \cdots [H_I(t_N), Q^I(t)] \cdots]] \rangle$$

- Work in a gauge in which the fluctuations vanish

Ag notes

- Gauge invariant perturbations at second order involve second order constraints in their definition.
- Gauge fixing strategy is more efficient and common in literature

Agullo notes

Non-gauss is determined by gravitational self interactions which become very strong in the plank regime.

Quantized perturbations living on a classical expanding background

Citations

- <https://arxiv.org/pdf/1905.05697.pdf> (plank group)

Example of a non-gaussian calculation

QFT and correlations (Power spectrum)

- Start with the action (AKA lagrangian). Min coupled to gravity
- Expand:
- Quantize perturbations
- Computer PS

QFT and correlations (Bispectrum)

- 3rd order to have interaction terms
- Bi spectrum
- Small order in slow roll
- Experimental finding

In-In formalism (Bispectrum)

- Different vacuums
- Three point correlation function
- Final form
- Experimental finding

- Third order in perturbations?
- In-In formalism to LQC. Curvature perturbations can only be treated at the end
- Non-Gaussianity of tensor perturbations in LQC
- The significance of using LQC is that it goes beyond providing linear perturbative corrections, how can non-pert. QG effects alter the metric pert.